Practice Questions

1. The demand curve for a product is given by Q = 300 – 2P+4I, where I=25 is the average income. The supply curve is Q = 3P – 50. Find the market clearing price and quantity.

   Hint:
   Set demand equal to the supply equation and substitute for I.
   P = 90
   Q = 220

2. A good is traded in the world market at Rs. 9. The domestic supply and demand for various price levels are given as:

Price	Supply	Demand
3	2	34
6	4	28
9	6	22
12	8	16
15	10	10

What will be the equation for demand?

Solution:

Q = a – bP

To find the slope b, find $\frac{\Delta Q}{\Delta P}$ = -6/3 = -2 = -b.

Now, to find a, substitute any price and quantity demanded (from table) to the equation Q = a – 2P.

34 = a – 2(3) implies a = 40.

Thus, demand equation will be Q = 40 – 2P.

Using same method, you can find supply equation also, just that general supply equation will be of the form Q = a + bP and use values from supply schedule of the table.

At price Rs. 9, what will be the price elasticity of demand?

Hint: Elasticity of demand = $\frac{\Delta Q}{\Delta P} * \frac{P}{Q} = \frac{(22-28)}{(9-6)} * \frac{9}{22} = 0.82$

3. The table below shows price of good x and good y for two years:

Year	Price of X	Sales of X	Price of Y	Sales of Y

2022	10.35	75	4.11	820
2023	10.48	70	3.76	850

Find the price elasticity of demand for good X. Derive the linear demand curve for good X.

Hint:

First estimate slope = $\frac{\Delta Q}{\Delta P} = \frac{850-820}{3.76-4.11}$ = -85.7

As the demand curve is linear, the elasticity will differ between two years. So, using slope, let us calculate elasticity for two years and find the arc elasticity at the average point between two years.

$$E_d^{2022} = \frac{\Delta Q}{\Delta P} * \frac{P}{Q} = \frac{4.11}{820} * (-85.7) = -0.043$$

$$E_d^{2023} = \frac{\Delta Q}{\Delta P} * \frac{P}{Q} = \frac{3.76}{850} * (-85.7) = -0.038$$

Arc elasticity

$$= \frac{\Delta Q}{\Delta P} * \frac{\bar{P}}{\bar{Q}} = \frac{3.935}{835} * (-85.7) = -0.040$$

To find the equation for demand curve, slope = -85.7

Q = a – 85.7P

To find a, substitute price and quantity values from the table.

820a – 85.7 * 4.11 implies a = 1172.2.

Substitute a and b to get the demand equation.

4. Given a consumer's utility function $U(X,Y) = XY$. The price of one unit of good X is Rs. 12 and price of one unit of good Y is Rs. 4. Suppose his budget is Rs. 48, find the utility maximising optimal bundle.

   Hint:
   Find MU of good x and MU of good y by differentiating the total utility function XY. Budget line equation will be 12X + 4Y = 48 or 3X + Y = 12. Slope will 12/4 = 3.

   Set Mux/MUy = 3 implies Y/X=3 implies any combination of goods where Y = 3X can be consumed provided the expenditure stays with the budget of Rs. 48 and gives maximum utility. Utility maximising bundle will be 6 units of Y and 2 units of X. Total expenditure = 2*12 + 6*4 = Rs. 48.